American Pie

Design Specification & Implementation Plan

Version 1.0 — March 2026 *A personal financial literacy simulator. Local. Private. Yours.*

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1. Project Overview

American Pie is a personal financial literacy simulator. It is not a product. It is not for market. It has one user: you.

The goal is not to build a finance app. The goal is for you to become financially literate and confident. The app is the mechanism. The knowledge is the outcome.

What It Is

A Wikipedia-style reading experience that slowly becomes a financial simulator without

announcing the transition. Articles are static until they are not. Components that look like illustrations come alive as you read. The app is always pulling you forward without ever pushing you through a wall of information.

Two Modes

Mode	What it is	Status
Simulation	Historical financial periods with mock data. The game.	Built first.
Real App	Your actual financial life with your real numbers.	Shell exists. One starter article authored.

The Game

The game is building accurate simulations. You play pre-built historical simulations to learn the variables. Then you research and build your own simulations for years you lived through. Your accuracy score improves as your financial understanding deepens. You eventually build forward-looking simulations for the current year. The only way to get better at the game is to become financially literate.

The Device Strategy

All data lives in a synced iCloud folder at `C:\iCloudDrive\american_pie`. The app reads and writes to this folder. Changes made on desktop appear on mobile and vice versa. No database. No server. No accounts. Just a folder of text files displayed beautifully.

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2. Guiding Principles

These principles govern every design and implementation decision. When in doubt, return here.

```
<details>
<summary><strong>P1 — The app is never empty</strong></summary>
```

From the first second the app loads, it shows real information. Before any data is entered it uses average American demographic data for someone in your approximate situation. The moment you enter one real number, that number becomes the seed of everything personal.

```
</details>
```

```
<details>
<summary><strong>P2 — The illusion is complete</strong></summary>
```

Every section of the app exists visually from day one. Undeveloped sections look identical to developed ones. They navigate correctly. They load correctly. They simply have no data behind them yet. The app looks finished. It is being filled in from the inside.

```
</details>
```

```
<details>
<summary><strong>P3 — Nothing is announced</strong></summary>
```

Components do not introduce themselves. The ticker appears as an illustration before it is live. The pressure gauge appears as a decorative element before it moves. The navigation appears after the first article is read, not before. The app reveals itself through use, not through onboarding.

```
</details>
```

```
<details>
<summary><strong>P4 — The clickbait is the curriculum</strong></summary>
```

Every hook, headline, and “did you know” item is chosen because it teaches a specific concept. The article it leads to is the lesson. The user never knows they are following a curriculum. They believe they are following their curiosity.

```
</details>
```

```
<details>
<summary><strong>P5 — No automation replaces your understanding</strong></summary>
```

The app does not fetch data for you. It does not read your documents. It does not make decisions. It tells you what to look for, where to find it, how to store it, and what it means once you have it. The manual process is the education.

```
</details>
```

```
<details>
<summary><strong>P6 — The simulation never saves</strong></summary>
```

When a simulation reaches the present day it resets. No portfolio to export. No score to keep.

The knowledge is the only thing that transfers. This removes the temptation to optimize the simulation instead of learning from it.

```
</details>


<details>
<summary><strong>P7 — Both themes maintain the serious illusion</strong></summary>
```

Neither theme looks like a web app. The Encyclopedia theme looks like Wikipedia. The Newspaper theme looks like a broadsheet. No rounded corners. No drop shadows. No buttons that look like buttons. The UI chrome is always invisible.

```
</details>


<details>
<summary><strong>P8 — The backend is always visible</strong></summary>
```

Source View is available on every screen. It shows exactly which file is powering what you see, what variables are in use, and how each calculation works. Learning the backend is not separate from using the app. It is the same activity.

```
</details>
```

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3. File Structure

```
C:\iCloudDrive\american_pie\
│
├── index.html                    ← The entire app. Open this in a browser.
├── style.css                     ← All visual styles. Both themes live here.
├── app.js                        ← All display and interaction logic.
│
├── articles\
│   ├── _template.md              ← Template for every new article.
│   ├── emergency-fund.md         ← Fully authored. The starter article.
│   ├── burn-rate.md
│   ├── checking-vs-savings.md
│   ├── federal-reserve.md
│   ├── bear-stearns.md
│   ├── mortgage-backed-securities.md
│   ├── unemployment-insurance.md
```

```
│   ├── income-tax-basics.md
│   ├── capital-gains.md
│   ├── index-funds.md
│   ├── compound-interest.md
│   ├── roth-ira.md
│   ├── inflation.md
│   ├── interest-rates.md
│   ├── net-worth.md
│   ├── credit-score.md
│   ├── dollar-cost-averaging.md
│   ├── sp500.md
│   ├── stimulus.md
│   ├── quantitative-easing.md
│   ├── bull-market.md
│   ├── lifestyle-inflation.md
│   ├── tax-brackets.md
│   ├── wash-sale.md
│   ├── sequence-of-returns.md
│   ├── i-bonds.md
│   ├── crypto-basics.md
│   └── [present]\
│       └── getting-started.md    ← The one authored real-app article.
│
├── simulations\
│   ├── 2008\
│   │   ├── variables.json        ← All historical data for 2008 simulation.
│   │   ├── events.json           ← Ordered event sequence for 2008.
│   │   ├── ghost-tracks.json     ← All five track logs for 2008.
│   │   └── did-you-know.json     ← All hooks for 2008 home page.
│   ├── 1999\
│   │   └── [placeholder files]
│   ├── 2020\
│   │   └── [placeholder files]
│   └── [present]\
│       ├── variables.json        ← Your real numbers. You fill this in.
│       ├── watch-list.json       ← What to monitor and how often.
│       └── did-you-know.json     ← Personal hooks from your real data.
│
├── worksheets\
│   ├── simulation-research.md    ← Printable variable research worksheet.
│   └── real-life-snapshot.md     ← Printable personal finance snapshot.
│
└── sources\
    ├── where-to-find-things.md   ← Master reference for every data source.
    └── [year]\
        └── [variable-name].md    ← Your source notes per variable per year.
```

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4. Data Architecture

Variables File

Every simulation has a `variables.json`. This is the single source of truth for all numbers in that simulation. The app reads from this file. The UI writes to this file. You can edit this file directly in any text editor.

```
{
  "fed_funds_rate": {
    "value": 4.25,
    "unit": "percent",
    "confirmed": true,
    "source": "federalreserve.gov",
    "last_checked": "2008-01-22",
    "notes": "Fed cut 75bps in emergency session January 22"
  },
  "your_monthly_expenses": {
    "value": 2100,
    "unit": "dollars",
    "confirmed": true,
    "source": "mock data",
    "last_checked": "2008-01-01",
    "notes": "Rent 950, food 400, car 300, other 450"
  },
  "your_cash": {
    "value": 7000,
    "unit": "dollars",
    "confirmed": true,
    "source": "mock data",
    "last_checked": "2008-01-01",
    "notes": "Starting position"
  }
}
```

Every field explained:

Field	What it means	Required
`value`	The number	Yes
`unit`	percent, dollars, days, etc.	Yes
`confirmed`	Is this verified data or an estimate?	Yes
`source`	Where you got this number	Yes
`last_checked`	When you last verified it	Yes

`notes`	Anything useful you want to remember	No

Articles File

Every article is a `.md` file. Plain text. Readable without the app. Variables are referenced with double brackets.

```
# Emergency Fund

An emergency fund is money set aside...

Your situation covers an estimated
**{{coverage_days}} days** of expenses.

This is calculated as:
{{your_cash}} ÷ {{your_monthly_expenses}} × 30
```

The app replaces `{{variable_name}}` with the value from `variables.json` before displaying.

Events File

The simulation event sequence. Each event is one entry. The app displays them in order and waits for user engagement before advancing.

```
[
  {
    "id": "2008-01",
    "date": "January 2008",
    "era": 1,
    "headline": "Fed cuts rates for the fourth time in six months",
    "ticker_text": "FED CUTS RATES 25BPS · FOURTH CUT IN SIX MONTHS",
    "track": "economy",
    "article": "federal-reserve",
    "variables_affected": ["fed_funds_rate"],
    "ghost_track_updates": {
      "winner": "Noticed savings account rate dropped. Moved to higher yield.",
      "weatherer": "Saw the news. Did not act.",
      "unaware": "Did not see the news.",
      "gambler_lost": "Increased short position in financials.",
      "gambler_won": "Increased short position in financials."
    }
  }
]
```

Ghost Tracks File

The full log for all five tracks across the entire simulation period.

```
{
  "winner": {
    "color": "#2D6A4F",
    "label": "The Winner",
    "philosophy": "Smart, boring, consistent decisions made early.",
    "log": [
      {
        "date": "January 2008",
        "cash": 7000,
        "monthly_burn": 2100,
        "coverage_days": 100,
        "credit_card_debt": 0,
        "net_worth": 7000,
        "action": "Calculated monthly burn rate for first time.",
        "thinking": "I need to know exactly where I stand."
      }
    ]
  }
}
```


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5. Themes

The theme is controlled by a single class on the `<body>` element. Either `theme-encyclopedia` or `theme-newspaper`. Switching is one line of JavaScript. All styling cascades from this class.

Theme 1 — Encyclopedia

```
<details>
<summary><strong>Typography</strong></summary>
```

Role	Font	Notes
Body text	Linux Libertine (Google Fonts)	Wikipedia's actual font
Data & numbers	IBM Plex Mono	Monospace. Terminal feel.
Headers	Linux Libertine, larger	Hierarchy through size only

```
</details>
```

```
<details>
```

<summary>Color Palette</summary>

Token	Hex	Use
`--bg`	`#F8F9FA`	Page background. Wikipedia grey.
`--text`	`#202122`	Body text. Wikipedia ink.
`--link`	`#3366CC`	All links. Wikipedia blue.
`--border`	`#A2A9B1`	Rules and dividers.
`--data`	`#000000`	Numbers and data values.
`--winner`	`#2D6A4F`	Ghost track: Winner
`--weatherer`	`#1D3557`	Ghost track: Weatherer
`--unaware`	`#E9C46A`	Ghost track: Unaware
`--gambler-lost`	`#E63946`	Ghost track: Gambler Lost
`--gambler-won`	`#6A994E`	Ghost track: Gambler Won

No other colors. No accent. No gradients. No shadows.

</details>

<details>
<summary>Layout Rules</summary>

- No rounded corners anywhere. `border-radius: 0` on everything.
- No box shadows anywhere.
- No buttons that look like buttons. All interactive elements are text links or inline fields.
- Thin `1px` borders only. Color `--border`.
- Two column: 65% content, 35% margin on desktop. Single column on mobile.
- Generous line height: `1.7` for body text.
- Compact line height: `1.3` for data values.

</details>

Theme 2 — Newspaper

<details>
<summary>Typography</summary>

Role	Font	Notes
Headlines	Playfair Display	Dramatic. High contrast serif.
Body text	Libre Baskerville	Dense newspaper column feel.
Data & bylines	Courier Prime	Typewriter. Dates, sources, numbers.
Pull quotes	Playfair Display Italic	Large. Grid-breaking.

</details>

<details>
<summary>Color Palette</summary>

Token	Hex	Use
`--bg`	`#F5F0E8`	Aged newsprint. Warm.
`--text`	`#1A1A1A`	Deep ink black.
`--headline`	`#0A0A0A`	Headlines heavier than body.
`--border`	`#2B2B2B`	Thick ink rules.
`--link`	`#1A1A1A`	Underlined. No blue. Editorial.
`--data`	`#8B0000`	Deep editorial red. Numbers only.
Ghost track colors	Desaturated versions of Encyclopedia palette	Same meaning, less digital feel.

</details>

<details>
<summary>Layout Rules</summary>

- Three narrow columns on desktop. Left: dates and markers. Center: article body. Right:

margin.

- Heavy horizontal rules between sections. `3px` solid `--border`.
- Dateline always visible at top in Courier Prime.
- Subtle paper grain texture via CSS `::before` pseudo-element. `opacity: 0.03`. Barely visible.
- Pull quotes break the column grid intentionally. Full center column width. Large Playfair Italic.
- No rounded corners. No shadows. No web-app feel.

```
</details>


<details>
<summary><strong>Newspaper-only Feature: The Dateline</strong></summary>
```

Always present at the very top. Full width. Courier Prime. Small caps.

```
AMERICAN PIE                    WEDNESDAY, MARCH 26, 2026
```

In simulation mode, shows the simulation date:

```
AMERICAN PIE                    TUESDAY, MARCH 11, 2008


</details>
```

Theme Switch

Located at the bottom of every screen. Looks like a Wikipedia language selector. No modal. No settings page.

```
Read in:  Encyclopedia · Newspaper
```

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---

6. Screen Specifications

6.1 Home Page

The first screen. Looks exactly like a Wikipedia main page. No app chrome. No navigation. No onboarding. Just a page.

```
<details>
<summary><strong>Masthead</strong></summary>


American Pie
The Free Financial Encyclopedia

____________________________________________________
```

Right-aligned on the same row as the title:

```
[ 2008 ]  [ Present · ● ]
```

The ● next to Present pulses at one breath per four seconds. Means the data is live. No tooltip. No label. You figure it out.

Font: Serif (Linux Libertine or Playfair Display depending on theme).

```
</details>


<details>
<summary><strong>Featured Article Block</strong></summary>
```

Full width. Light left border (grey in Encyclopedia, thick black rule in Newspaper).

```
★ Featured Article

____________________________________________________

The 2008 Financial Crisis: What Most Americans Did Not
Know Was Already Happening To Them

By late 2007, the signals were already present. Housing
prices had peaked. Credit was overextended. The
institutions most Americans trusted with their money
were quietly insolvent.

Most households would not understand what had happened
until it was already over. By then, the cost of not
knowing had already been paid.

                                              Read more →
____________________________________________________
```

`Read more →` **is not a button. It is a text link. The only interactive element in this block.**

```
</details>


<details>
<summary><strong>Left Column — In The News</strong></summary>


In the news — Early 2008
```

```
· Bear Stearns reports $1.9B loss tied to
  mortgage-backed securities

· Federal Reserve cuts interest rates for the
  fourth time in six months

· US unemployment reaches 5.0%, highest since 2004

· Median home prices fall for first time since
  Great Depression

· Oil prices reach $100 per barrel for first
  time in history

· Consumer confidence index falls to 16-year low

· IndyMac Bank seized by federal regulators
```

Each item: link, `--link` color, underline on hover only. Each opens a different article. Each teaches a different concept.

```
</details>


<details>
<summary><strong>Right Column — Did You Know</strong></summary>


Did you know?
```

```
· The average American household lost 39% of
  their net worth between 2007 and 2010.

· 57% of Americans had no emergency fund in 2008.

· A person who invested $7,000 in March 2009
```

```
  would have $47,000 by 2024.

· Most people learned what a mortgage-backed
  security was only after losing money
  because of one.
```

The `$7,000` figure is seeded from the mock data. It is your number. The app does not announce this.

Once you enter your real cash amount in any article sidebar, all instances of this figure update throughout the app.

```
</details>


<details>
<summary><strong>Bottom Strip — On This Day</strong></summary>


─────────────────────────────────────────────
On this day — March 14, 2008

JP Morgan acquired Bear Stearns for $2 per share.
One year earlier the stock traded at $172.

People who understood what this meant acted within hours.
─────────────────────────────────────────────
```

No link. No interaction. Rotates from `did-you-know.json` each session.

```
</details>
```

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6.2 Article View

The primary reading screen. Every concept, every simulation event lives here. The structure is always identical. The content changes.

```
<details>
<summary><strong>Layout</strong></summary>
```

- Desktop: 65% content left column, 35% margin right column.
- Mobile: single column. Margin collapses to expandable panel at bottom.

- **No scrolling within a section. Each section is one screen tall. You advance with `Next →`.**
- **The section never moves. Only the content changes. Slideshow mechanic.**

```
</details>


<details>
<summary><strong>Article Header</strong></summary>


━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
American Pie  ›  2008  ›  Banking

Emergency Fund
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━

March 2008  ·  6 sections  ·  Concepts: 3
```

Breadcrumb is navigable. Each crumb is a link back to that level.

```
</details>


<details>
<summary><strong>Section Structure — Content Column</strong></summary>
```

Every section follows this pattern:

```
Section N of 6

[Plain language explanation. Two to four sentences.
Never more. The explanation is complete on its own.]

[Your number, displayed inline. If it references a
variable, the variable value appears here.]

Your situation: ▸ [editable field] days

━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
                                            Next →
```

Section types and their content rules:

Section	Name	What it contains
1	What it is	Plain language definition. No jargon.

2	Your number	One calculation using your variables. Editable inline.
3	The math	The calculation shown step by step using your actual values.
4	What people did	What most people did. What smart people did. Reveal for “if you’d known earlier.”
5	What it connects to	Three tappable chips linking to related concepts.
6	Your next step	One concrete action. Specific. Doable today.

```
</details>


<details>
<summary><strong>Section 4 — What People Did (Detail)</strong></summary>


Section 4 of 6

What most people did

In 2008, 61% of Americans who lost their job waited
more than 3 weeks to reduce spending. By then, the
average household had used $1,400 of savings
unnecessarily.

— — — — — — — — — — — — — — — — — — — — — — — —

What financially stable people did

They calculated their exact burn rate within 48 hours
of job loss and cut non-essential spending immediately.
Not because they panicked. Because the math was clear.

— — — — — — — — — — — — — — — — — — — — — — — —

If you had known this in 2007

  ▸ Reveal

                                                Next →
```

▸ `Reveal` **expands one paragraph. Personal. Slightly uncomfortable. Collapses when you advance.**

```
</details>
```

```
<details>
<summary><strong>The Margin Panel — Right Column</strong></summary>
```

Always visible. Never intrusive.

Default state (nothing active):

```
────────────────
  In this article
────────────────

  1 · What it is
  2 · Your number
  3 · The math
  4 · What people did
  5 · What it connects to
  6 · Your next step

────────────────
  Ghost Tracks
────────────────

  🟢 Winner
  🔵 Weatherer
  🟡 Unaware
  🔴 Gambler−
  ⚪ Gambler+

  Tap any track
────────────────
```

Active state (after tapping inline editable field):

```
────────────────
  Your number
────────────────

  Monthly expenses
  [ $_______ ]

  Current savings
  [ $_______ ]

  ────────────────

  Using mock data:
  Expenses: $2,100
  Savings:  $7,000
```

```
Coverage: 100 days

────────────────

Enter your real
numbers above to
see your actual
coverage.

────────────────
```

Closes when you tap anywhere in the article body.

Ghost track expanded state:

```
────────────────
🟡 Unaware
March 2008
────────────────

Cash:       $4,200
Burn/mo:    $2,100
Coverage:   60 days

Credit card: $800

Last action:
Paid minimum on
credit card. Did
not calculate
burn rate.

────────────────

What they're
thinking:
"Things will
pick up soon."

────────────────
```

</details>

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6.3 Ghost Track Full View

Accessible by tapping any ghost track name. Opens as a full-page overlay sliding from the right. Looks like a Wikipedia article about a person.

```
<details>
<summary><strong>Layout</strong></summary>


────────────────────────────────────────────────
American Pie  ›  2008  ›  Profiles

🟢 The Winner
────────────────────────────────────────────────

March 2008  ·  Age 28  ·  Renting
Income: $2,100/mo  ·  Cash: $7,000
────────────────────────────────────────────────
```

Below the header: a running log. Date left. Entry right. Like a ship's log.

```
Jan 2008   Calculated monthly burn rate for the
           first time. Result: $2,100/month.

Jan 2008   Opened high-yield savings account.
           Moved $4,200 (2 months expenses).

Feb 2008   Noticed hours may be reduced. Cut
           3 subscriptions. Saved $87/month.

Mar 2008   Bear Stearns collapsed. Did nothing.
           Already prepared.


</details>


<details>
<summary><strong>Final Status Screen (End of Simulation)</strong></summary>
```

Each track ends with a summary:

```
────────────────────────────────────────────────
🟢 The Winner — Final Status: 2026
────────────────────────────────────────────────

Started:      $7,000 cash · No investments
Ended:        $340,000 net worth
```

```
Key decisions: 4
Luck required: None
Knowledge required: Moderate

Philosophy:   Boring is powerful.
```

```
</details>
```

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6.4 Clarity Score

Accessible from the persistent small number in the top right of every screen after the home page.

```
<details>
<summary><strong>Layout</strong></summary>
```

```
Your Financial Clarity

  Score: 23 / 100

  You understand 23% of what affects
  your financial life.

  That is 23% more than when you started.

  What you understand

  Emergency funds          ▓▓▓▓▒▒▒▒
  Burn rate                ▓▓▓▓▓▓▒▒
  Checking vs savings      ▓▓▓▓▒▒▒▒
  Federal Reserve          ▓▓▒▒▒▒▒▒
  Income tax basics        ▒▒▒▒▒▒▒▒
  Investment accounts      ▒▒▒▒▒▒▒▒
  Capital gains            ▒▒▒▒▒▒▒▒
  Inflation                ▒▒▒▒▒▒▒▒

  Next concept that will move your
```

```
score the most:

→ Burn Rate (you are close)
```

Bars use block characters. Monospace. Terminal feel. Empty bars are not locked. Just empty. Waiting.

```
</details>
```

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6.5 Navigation Shell

Appears after the first complete article is read. Not before.

```
<details>
<summary><strong>Desktop Sidebar</strong></summary>


╔═══╗
║ ⌂ ╒  Home
╞═══╡
║ ◎ ╒  Simulation
╞═══╡
║ ≡ ╒  My Situation
╞═══╡
║ ∞ ╒  Learn
╞═══╡
║ △ ╒  Clarity  23
╚═══╝
```

Labels appear on hover. Clarity score lives next to the icon permanently.

```
</details>


<details>
<summary><strong>Mobile Bottom Bar</strong></summary>
```

Five icons. Full width. Same icons. Labels always visible on mobile since hover doesn't exist.

```
</details>


<details>
<summary><strong>Undeveloped Section Behavior</strong></summary>
```

Undeveloped sections navigate correctly. They load correctly. The header loads. The breadcrumb loads. The margin loads. Then where content would be:

```
This section has no data yet.

Continue in simulation to build context →
```

Not “coming soon.” Not locked. Just honest. Always points somewhere that is built.

```
</details>
```

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---

7. Core Components

7.1 Ticker Bar

```
<details>
<summary><strong>Behavior and First Appearance</strong></summary>
```

First appears in the article explaining financial news tickers. It looks like a screenshot — a flat illustration embedded in the article.

```
────────────────────────────────────────────────────
Example: How markets display real-time information

 DOW  -504  ▼    S&P  -106  ▼    OIL  $101  ▲
────────────────────────────────────────────────────
```

Two sections later, the numbers have changed. You didn’t click anything. From this point it is always present at the bottom of every screen. It never announces itself.

```
</details>


<details>
<summary><strong>Content</strong></summary>
```

Five color-coded tracks cycling continuously. One update per track every 30 seconds. Updates come from `ghost-tracks.json` current entry for the active simulation date.

```
🟢 Winner · Just moved 3 months expenses to high-yield savings
```

```
🔵 Weatherer · Cut subscriptions, extended runway 6 weeks
🟡 Unaware · Withdrew $2,000 to cover rent
🔴 Gambler– · Shorted Lehman. Position up 40%.
⚪ Gambler+ · Shorted Lehman. Position up 40%.
```

Note: Gambler– and Gambler+ are identical right now. They diverge later. This is intentional tension.

```
</details>


<details>
<summary><strong>Visual Spec</strong></summary>
```

- **Full width. Fixed at bottom of every screen above theme switch.**
- **Height:** `32px`**.**
- **Background:** `--text` **color (inverted from page).**
- **Text:** `--bg` **color (inverted from page).**
- **Font: Monospace.** `11px`**.**
- **Scrolls horizontally. Seamless loop. CSS animation only. No JavaScript required for the scroll.**
- **Color dots before each entry match ghost track color.**

```
</details>
```

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7.2 Pressure Gauge

```
<details>
<summary><strong>First Appearance</strong></summary>
```

Appears in the Federal Reserve article as a decorative illustration. Looks static. Is not static.

```
Economic Stress Indicator — Early 2008
[ ▸━━━
```

is present above the ticker on every screen.

```
</details>


<details>
<summary><strong>Data Source</strong></summary>
```

In simulation: reads from `events.json`. Each era has a `stress_level` value from `0` to `100`. The gauge moves to match the current era's stress level over time.

In real app: calculates from your personal variables — burn rate days remaining, debt-to-income ratio, income stability flag, and current Fed stress index from your manually entered data.

```
</details>


<details>
<summary><strong>Visual Spec</strong></summary>
```

- Full width horizontal bar above ticker.
- Height: `6px`.
- Background gradient: `#4A90D9` (cool blue) to `#E63946` (crisis red).
- Marker: `10px × 10px` square. Color `--text`. Positioned via CSS `left` percentage.
- Transitions: `left` property transitions over `5000ms ease-in-out`. Barely perceptible movement.
- No labels visible unless hovered. On hover: `Calm` left, `Crisis` right. `10px` monospace text.

```
</details>
```

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7.3 Ghost Tracks Margin Panel

Always visible in the right margin of every article. Described fully in Section 6.2.

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7.4 Inline Variable Editor

```
<details>
<summary><strong>Trigger</strong></summary>
```

Any `{{variable_name}}` in article text renders with a subtle ▸ indicator after it. Tapping the value or the indicator opens the margin panel in edit mode.

```
</details>
```

```
<details>
<summary><strong>Behavior</strong></summary>
```

1. Margin panel opens with edit fields for relevant variables
2. Mock data shown as placeholder values
3. User types real values
4. App writes to `variables.json` immediately on input
5. Article updates inline — the number changes in the text without page reload
6. Margin closes on next tap in article body

```
</details>
```

```
<details>
<summary><strong>Visual Spec</strong></summary>
```

- Editable values in article body: monospace, `--data` color, subtle underline dotted.
- Input fields in margin: plain `1px --border` border. No styling. Feels like a form field from 2005. Intentional.
- No save button. Writes on input event. No confirmation needed.

```
</details>
```

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7.5 Source View

```
<details>
<summary><strong>Access</strong></summary>
```

Available on every screen after the home page. Small < > link in the breadcrumb row. Tapping it opens an overlay panel from the bottom.

```
</details>


<details>
<summary><strong>Content</strong></summary>


━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
Source View
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━

This screen is powered by:

Article file:
articles/emergency-fund.md

Variables used:
your_monthly_expenses  →  $2,100
your_savings           →  $7,000
coverage_days          →  100

coverage_days is calculated:
your_savings ÷ your_monthly_expenses × 30
= 7,000 ÷ 2,100 × 30
= 100 days

To edit this article:
Open articles/emergency-fund.md in any text editor.

To edit these variables:
Open simulations/2008/variables.json
or tap any value in the article to edit in the app.

━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━


</details>
```

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8. The Five Ghost Tracks

All five start in January 2008 with identical circumstances: age 28, renting, $7,000 cash, $2,100/month expenses, no investments, employed.

Track	Color	Label	Philosophy	2026 Net Worth
🟢	`#2D6A4F`	The Winner	Smart, boring, consistent decisions made early.	~$340,000
🔵	`#1D3557`	The Weatherer	Never got ahead. Never collapsed. Low stress, no regrets.	~$85,000
🟡	`#C9952A`	The Unaware	Did not know the rules. Most common outcome.	~$12,000
🔴	`#C1121F`	The Gambler Who Lost	Knew everything. Acted on it badly.	~-$40,000
⚪	`#6A994E`	The Gambler Who Won	Same knowledge. Better timing. Admits luck played a role.	~$1,200,000

The five tracks teach one lesson: identical starting points produce radically different outcomes based on decisions, knowledge, and variance. The Gambler Who Lost and the Gambler Who Won made the same kinds of decisions. One detail — leverage — separated them.

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---

9. Simulation Arc — 2008 to 2026

Era 1 — The Calm Before (Late 2007 – Early 2008)

```
<details>
<summary><strong>World context</strong></summary>
```

Everything looks fine. Jobs exist. People are buying houses. Credit is easy. The stock market peaked October 2007 and has started quietly declining but nobody is panicking. The word recession has not been used officially.

```
</details>


<details>
<summary><strong>Your mock character</strong></summary>
```

Employed. Renting. $7,000 in a checking account earning almost nothing. No investments. No

debt besides a car payment. Life feels normal and slightly precarious.

```
</details>


<details>
<summary><strong>Concepts taught</strong></summary>
```

- What a checking account actually costs you in lost opportunity
- What interest rates are and why the Fed keeps cutting them
- What inflation is doing to your $7,000 right now even when nothing feels wrong
- What a credit score is and why yours matters more than you think
- What “the market peaked” means for someone who owns nothing

```
</details>


<details>
<summary><strong>Key ghost track divergence point</strong></summary>
```

The Winner moves 2 months of expenses into a high-yield savings account. One small action. No one else does this. This single decision adds $840 in interest over the next 18 months and establishes the habit of thinking about where money sits.

```
</details>


<details>
<summary><strong>Clickbait hooks</strong></summary>


Why your savings account is making someone else rich

The number your bank does not want you to calculate

What "the Fed cut rates again" actually did to your
grocery bill

5 things that were already broken in 2007 that
nobody noticed


</details>
```

```
<details>
<summary><strong>Era-ending unresolved thread</strong></summary>
```

A “did you know” item surfaces: Bear Stearns employees lost their entire retirement savings in one week. No explanation of what Bear Stearns is. The link is there. The next era begins.

```
</details>
```

```
<details>
<summary><strong>Stress level</strong></summary>
```

12 / 100

```
</details>
```

Era 2 — The Crack (March 2008 – September 2008)

```
<details>
<summary><strong>World context</strong></summary>
```

Bear Stearns collapses March 14. The Fed brokers an emergency sale. Everyone says it is contained. It is not contained. Housing prices are falling nationally for the first time since the Great Depression. Banks are quietly terrified of each other. Nobody says recession officially yet.

```
</details>
```

```
<details>
<summary><strong>Your mock character</strong></summary>
```

Still employed but hours quietly reduced. The $7,000 is still in checking. You have not noticed anything dramatic yet. But your monthly surplus is shrinking.

```
</details>
```

```
<details>
<summary><strong>Concepts taught</strong></summary>
```

- What Bear Stearns was and why it mattered to people who never heard of it

- What a mortgage-backed security is in plain language
- What the Federal Reserve actually does and why it matters to you
- What a recession is technically versus what it feels like
- What reduced hours does to your annual tax situation
- What an emergency fund is and how to calculate yours specifically
- The difference between a checking account, savings account, and money market

```
</details>


<details>
<summary><strong>Key ghost track divergence point</strong></summary>
```

The Winner calculates their burn rate for the first time. This moment — the first time someone actually knows their number — is the single most important financial action of the entire simulation. Everything downstream depends on knowing this number.

```
</details>


<details>
<summary><strong>Clickbait hooks</strong></summary>


Bear Stearns had 14,000 employees on a Friday.
By Monday it was gone.

What "contained" means when a banker says it

Your paycheck got smaller. Here is the thing
nobody told you.

How to calculate exactly how many days your
savings will last


</details>


<details>
<summary><strong>Era-ending unresolved thread</strong></summary>
```

Article about emergency funds ends with one line: “Lehman Brothers has announced it is exploring strategic alternatives.” No explanation. A link. You must follow it.

```
</details>


<details>
<summary><strong>Stress level</strong></summary>
```

38 / 100

```
</details>
```

Era 3 — The Fall (September 2008 – December 2008)

```
<details>
<summary><strong>World context</strong></summary>
```

Lehman Brothers files bankruptcy September 15. Largest bankruptcy in American history. The stock market loses 777 points in one day. Credit markets freeze. Congress passes a $700 billion bailout. Nobody can explain it in plain language.

```
</details>


<details>
<summary><strong>Your mock character</strong></summary>
```

Laid off. This is where the simulation mirrors your real life most directly. You now have $7,000, no income, and a countdown clock. Every decision matters more than it ever has.

```
</details>


<details>
<summary><strong>Concepts taught</strong></summary>
```

- What happens to your taxes when you lose your job mid-year
- What unemployment insurance is, how to apply, and what it actually pays
- What COBRA health insurance is and whether you can afford it
- What your actual monthly burn rate is with precision
- What the $700 billion bailout was and why people were furious
- What “too big to fail” means and what it costs taxpayers

- **What deflation is and why it is dangerous**
- **What a bear market is and what it does to people who panic sell**

```
</details>


<details>
<summary><strong>Key ghost track divergence point</strong></summary>
```

The Unaware does not file for unemployment immediately. Feels embarrassing. This delays their first check by 3 weeks and costs $1,260 in lost benefits. The Unaware also begins spending savings without a plan. Credit card balance reaches $3,000.

```
</details>


<details>
<summary><strong>Clickbait hooks</strong></summary>


Lehman Brothers lasted 158 years.
It was gone by lunch.

What $700 billion actually bought and who got it

The embarrassing thing 47% of newly unemployed
people do not do that costs them thousands

Your $7,000 has an expiration date. Here is the
exact number of days.


</details>


<details>
<summary><strong>Era-ending unresolved thread</strong></summary>
```

The Gambler Who Won track quietly buys something in December 2008. The ticker just says: `Gambler+ · Bought.` **No details. You must follow them into Era 4.**

```
</details>


<details>
<summary><strong>Stress level</strong></summary>
```

```
</details>
```

Era 4 — The Bottom (January 2009 – March 2009)

```
<details>
<summary><strong>World context</strong></summary>
```

The market continues falling. March 9 is the actual bottom — S&P 500 closes at 676. Nobody knows it is the bottom at the time. Obama takes office. Stimulus bill passes. Every instinct says something terrible is still coming.

```
</details>
```

```
<details>
<summary><strong>Your mock character</strong></summary>
```

Unemployment checks arriving. Burn rate managed but tight. $7,000 slowly becoming $5,000. Surviving but not comfortable. The news is relentlessly dark.

```
</details>
```

```
<details>
<summary><strong>Concepts taught</strong></summary>
```

- What the stimulus check was designed to do
- What “buying the dip” means and why it is psychologically almost impossible in real time
- What dollar cost averaging is
- What an index fund is in plain language
- What the S&P 500 actually is and why it matters to someone with no investments
- What opportunity cost is — specifically what not investing $1,000 in March 2009 cost over 15 years
- What fear does to financial decision making

```
</details>
```

```
<details>
<summary><strong>Key ghost track divergence point</strong></summary>
```

The Winner uses $1,000 of savings to buy an S&P 500 index fund on March 9, 2009. Terrifying at the time. The math said to. The December 2008 purchase by Gambler Who Won is revealed: they bought the index at 850. It is now at 676. They are down 20%. They do not sell.

```
</details>


<details>
<summary><strong>Clickbait hooks</strong></summary>


The single best day to invest in the last 50 years
was the day it felt most insane to do it

What $1,000 invested on March 9, 2009 is worth today

The 401k mistake that cost Americans $300 billion
in penalties

Why your brain is specifically designed to make
you lose money


</details>


<details>
<summary><strong>Era-ending unresolved thread</strong></summary>
```

A quiet note appears in the Winner track: “Applied for one job today.” Recovery is beginning somewhere. You do not know where yet.

```
</details>


<details>
<summary><strong>Stress level</strong></summary>
```

71 / 100

```
</details>
```

Era 5 — The Long Climb (2009 – 2013)

```
<details>
<summary><strong>World context</strong></summary>
```

The market recovers slowly then faster. Jobs return unevenly. The word “recovery” is used before most people feel recovered. Housing stays depressed. A new normal settles.

```
</details>
```

```
<details>
<summary><strong>Concepts taught</strong></summary>
```

- What rebuilding an emergency fund looks like month by month
- What compound interest actually does with real numbers over time
- What a Roth IRA is and why starting one now matters
- What employer 401k matching is — free money most people refuse
- What net worth is and how to calculate yours
- The difference between assets and liabilities in plain language
- What a budget actually is versus what people think it is

```
</details>
```

```
<details>
<summary><strong>Stress level</strong></summary>
```

34 / 100

```
</details>
```

Era 6 — The Bull Market Nobody Trusted (2013 – 2019)

```
<details>
<summary><strong>World context</strong></summary>
```

Longest bull market in American history. Stocks rise almost every year. People who stayed invested get quietly wealthy. People who stayed in cash quietly fall behind. Bitcoin becomes

cultural. Gig economy emerges. Wealth inequality widens.

```
</details>
```

```
<details>
<summary><strong>Concepts taught</strong></summary>
```

- What a bull market is and why they end
- What inflation does to idle cash over 6 years
- What Bitcoin is, what it is not, and its actual risk profile
- What lifestyle inflation is and how it silently erodes wealth
- What tax brackets actually are — the raise-cost-you-money misconception
- What diversification means in practice
- What sequence of returns risk is
- What the gig economy does to your taxes

```
</details>
```

```
<details>
<summary><strong>Era-ending unresolved thread</strong></summary>
```

December 2019. One line in the news feed: “A novel pneumonia-like illness has been identified in Wuhan, China.” No article. No link. Just the date and the sentence.

```
</details>
```

```
<details>
<summary><strong>Stress level</strong></summary>
```

18 / 100

```
</details>
```

Era 7 — The Speed Crash (February 2020 – April 2020)

```
<details>
```

```
<summary><strong>World context</strong></summary>
```

Fastest stock market crash in history. 34% drop in 33 days. Then fastest recovery in history. Stimulus checks. PPP loans. Eviction moratoriums. The Fed prints money at unprecedented scale.

```
</details>
```

```
<details>
<summary><strong>Concepts taught</strong></summary>
```

- **What a market correction versus a crash actually is**
- **What quantitative easing is in plain language**
- **What stimulus money is designed to do and what it actually did**
- **What eviction moratoriums are and who they helped**
- **What happens to investments if you panic sell in a crash**

```
</details>
```

```
<details>
<summary><strong>Key ghost track divergence point</strong></summary>
```

The Unaware sells everything in March 2020 at the bottom. Locks in losses permanently. Misses the recovery entirely. The Winner adds to positions on the way down.

```
</details>
```

```
<details>
<summary><strong>Era-ending unresolved thread</strong></summary>
```

April 2020. Markets recovering. Article note: “Lumber prices up 300%. Used cars up 40%. Something is happening to prices.” The word inflation appears without explanation.

```
</details>
```

```
<details>
<summary><strong>Stress level</strong></summary>
```

82 / 100 at crash peak · 29 / 100 by April

```
</details>
```

Era 8 — Inflation (2021 – 2022)

```
<details>
<summary><strong>World context</strong></summary>
```

Inflation hits 9.1% in June 2022. Highest in 40 years. The Fed raises rates 11 times. Housing becomes unaffordable for most first-time buyers. Everything costs more. Wages rise but not as fast as prices. Crypto collapses. Tech stocks crash.

```
</details>
```

```
<details>
<summary><strong>Concepts taught</strong></summary>
```

- What inflation actually is mechanically
- What the Consumer Price Index is and how it is calculated
- What the Fed funds rate is and what raising it does to your life
- What happens to bond prices when interest rates rise
- What a Series I savings bond is and why it briefly mattered
- What real returns versus nominal returns means
- What housing affordability means mathematically
- What a wage-price spiral is

```
</details>
```

```
<details>
<summary><strong>Stress level</strong></summary>
```

61 / 100

```
</details>
```

Era 9 — The AI Era and The Present (2023 – 2026)

```
<details>
<summary><strong>World context</strong></summary>
```

Inflation cools but prices do not fall. AI reshapes industries and job markets. Stock market recovers led by tech. Interest rates stay high. Housing remains unaffordable. A new economic anxiety — not recession fear but structural uncertainty about what work looks like in five years.

```
</details>
```

```
<details>
<summary><strong>Concepts taught</strong></summary>
```

- What AI means for employment and which skills remain valuable
- What high interest rates mean for someone who wants to buy a house
- What current stock market valuation means
- What your actual next steps are given everything learned
- How to read your own financial documents now that you know what to look for
- What the real app needs to do for you specifically

```
</details>
```

```
<details>
<summary><strong>Ghost track final outcomes</strong></summary>
```

Track	Started	Ended	Key lesson
🟢 Winner	$7,000	~$340,000	Boring is powerful
🔵 Weatherer	$7,000	~$85,000	Stability is valid
🟡 Unaware	$7,000	~$12,000	The most common outcome
🔴 Gambler Lost	$7,000	~-$40,000	Knowledge does not eliminate risk
⚪ Gambler Won	$7,000	~$1,200,000	Luck played a role. They admit this.

```
</details>
```

```
<details>
<summary><strong>The final screen</strong></summary>


━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
The simulation is complete.

You started in 2008 with $7,000 and no financial
knowledge.

You now know what each decision costs and what
it returns.

The real app is waiting.
Your numbers are different.
The concepts are the same.

[ Start over ]          [ Switch to Present ]
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
```

No fanfare. No score. Just a choice.

```
</details>
```

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---

10. Full Article Manifest — 2008 Simulation

Every article required to fully author the 2008 simulation. Status tracked here.

#	Article File	Era	Concept	Status
1	`emergency-fund.md`	1–3	What an emergency fund is and how to calculate yours	✅ Fully authored
2	`burn-rate.md`	2	Monthly burn rate — what it is and how to calculate precisely	📝 To author
3	`checking-vs-savings.md`	1	The real cost of keeping money in checking	📝 To author
4	`federal-reserve.md`	1–2	What the Fed is, what it does, and why it matters to you	📝 To author
5	`bear-stearns.md`	2	What Bear Stearns was and why it	📝 To

			mattered to non-investors	author
6	`mortgage-backed-securities.md`	2	What an MBS is in plain language	📝 To author
7	`recession-defined.md`	2–3	What a recession is technically versus what it feels like	📝 To author
8	`unemployment-insurance.md`	3	What UI is, how to apply, what it pays	📝 To author
9	`cobra-insurance.md`	3	What COBRA is and how to decide if you can afford it	📝 To author
10	`taxes-when-laid-off.md`	3	What job loss does to your tax situation mid-year	📝 To author
11	`too-big-to-fail.md`	3	What this phrase means and what it cost taxpayers	📝 To author
12	`bear-market.md`	3	What a bear market is and what panic selling actually costs	📝 To author
13	`stimulus.md`	4	What stimulus checks are designed to do	📝 To author
14	`buying-the-dip.md`	4	What this means and why it is psychologically almost impossible	📝 To author
15	`dollar-cost-averaging.md`	4	What DCA is and how it removes timing pressure	📝 To author
16	`index-funds.md`	4–5	What an index fund is in plain language	📝 To author
17	`sp500.md`	4	What the S&P 500 actually is	📝 To author
18	`opportunity-cost.md`	4	What not investing in March 2009 actually cost over 15 years	📝 To author
19	`compound-interest.md`	5	What compound interest does with real numbers over time	📝 To author
20	`roth-ira.md`	5	What a Roth IRA is and why starting one now matters	📝 To author
21	`401k-matching.md`	5	What employer matching is — the free money most people refuse	📝 To author

22	`net-worth.md`	5	What net worth is and how to calculate yours	📝 To author
23	`assets-vs-liabilities.md`	5	The difference in plain language	📝 To author
24	`budget-reality.md`	5	What a budget actually is versus what people think it is	📝 To author
25	`bull-market.md`	6	What a bull market is and why they end	📝 To author
26	`idle-cash-inflation.md`	6	What inflation does to cash sitting for 6 years	📝 To author
27	`crypto-basics.md`	6	What Bitcoin is, what it is not, its actual risk profile	📝 To author
28	`lifestyle-inflation.md`	6	How rising income silently erodes wealth if unchecked	📝 To author
29	`tax-brackets.md`	6	What brackets actually are — the raise misconception	📝 To author
30	`diversification.md`	6	What diversification means in practice	📝 To author
31	`sequence-of-returns.md`	6	What sequence of returns risk is	📝 To author
32	`gig-economy-taxes.md`	6	What gig work does to your taxes specifically	📝 To author
33	`market-crash-vs-correction.md`	7	The technical difference and what it means for decisions	📝 To author
34	`quantitative-easing.md`	7	What QE is in plain language	📝 To author
35	`eviction-moratorium.md`	7	What moratoriums are and who they helped	📝 To author
36	`panic-selling-cost.md`	7	What selling at the bottom actually costs long term	📝 To author
37	`inflation-mechanics.md`	8	What inflation actually is — not just “prices go up”	📝 To author

38	`cpi.md`	8	What the Consumer Price Index is and how it is calculated	📝 To author
39	`fed-funds-rate.md`	8	What raising the rate does to your life specifically	📝 To author
40	`bonds-and-rates.md`	8	What happens to bond prices when rates rise	📝 To author
41	`i-bonds.md`	8	What Series I savings bonds are and when they matter	📝 To author
42	`real-vs-nominal-returns.md`	8	The difference and why it matters to every investment	📝 To author
43	`housing-affordability.md`	8	What affordability actually means mathematically	📝 To author
44	`wage-price-spiral.md`	8	What this is and how to recognize it	📝 To author
45	`ai-and-employment.md`	9	What AI means for jobs and which skills remain valuable	📝 To author
46	`high-rates-home-buying.md`	9	What current rates mean for a first-time buyer	📝 To author
47	`market-valuation.md`	9	What current valuations mean for someone starting to invest	📝 To author
48	`reading-your-documents.md`	9	How to read your own financial documents	📝 To author

Real App starter article:

#	Article File	Status
49	`[present]/getting-started.md`	✅ Fully authored

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11. Variable Manifest

Every variable used anywhere in the app. All live in `variables.json` for their respective simulation. In the real app they live in `[present]/variables.json`.

Variable Key	Unit	Description	Where to Find It	How Often Changes
`your_cash`	dollars	Your total liquid cash	Bank statement	Monthly
`your_monthly_expenses`	dollars	Average monthly spending	Last 3 bank statements ÷ 3	Monthly
`your_monthly_income`	dollars	Take-home pay per month	Pay stub (net amount)	When job changes
`coverage_days`	days	Calculated: `your_cash ÷ your_monthly_expenses × 30`	Calculated	Auto
`fed_funds_rate`	percent	Federal funds target rate	federalreserve.gov	When Fed meets
`inflation_rate`	percent	Current CPI annual rate	bls.gov/cpi	Monthly
`unemployment_rate`	percent	National unemployment rate	bls.gov	Monthly
`sp500_level`	points	S&P 500 index value	finance.yahoo.com	When relevant
`your_credit_card_debt`	dollars	Total credit card balance	Credit card statement	Monthly
`your_credit_score`	number	Current credit score	Free via Credit Karma	Quarterly
`your_net_worth`	dollars	Calculated: assets minus liabilities	Calculated	Monthly
`your_investment_value`	dollars	Total value of all investments	Brokerage statement	Monthly
`your_monthly_rent`	dollars	Monthly rent payment	Lease	When lease changes
`marginal_tax_rate`	percent	Your federal marginal bracket	irs.gov tax tables	Annually
		Short-term capital	Equals marginal	

short_term_cg_rate	percent	gains rate	rate	Annually
long_term_cg_rate	percent	Long-term capital gains rate	irs.gov	Annually
high_yield_savings_rate	percent	Current HYSA rate	Your bank's current rate	When rates change
stress_level	0–100	Economic or personal stress. Drives pressure gauge.	Calculated	Per era / monthly

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12. Real App Translation Map

Every simulation component and how it maps to the real app. The visual and structural design never changes. Only the data source changes.

Simulation	Real App
Featured article selected by era	Featured article selected by most pressing real financial situation
"In The News" — historical 2008 headlines	"In The News" — real current financial headlines you manually curate
"Did You Know" — average American data seeded with mock $7,000	"Did You Know" — recalculated entirely around your real entered numbers
"On This Day" — historical event from simulation date	"On This Day" — real historical financial event from today's actual date
Pressure gauge — simulated economic stress by era	Pressure gauge — your personal financial stress score from real variables
Ticker — ghost track log entries	Ticker — your own recent financial events and current watch list items
Ghost tracks — five fictional characters	Ghost tracks — composite anonymized profiles for your income bracket
Ghost track log — pre-written simulation entries	Ghost track log — your own log built entry by entry as you use the app

Inline variable editor — writes to simulation variables.json	Inline variable editor — writes to [present]/variables.json
Source view — shows simulation file paths	Source view — shows real file paths and your actual calculations
Year filter shows 2008 active	Year filter shows Present active with live pulse indicator
Clarity score — measures simulation concept engagement	Clarity score — measures understanding AND real-life application
Article "Your number" — uses mock data	Article "Your number" — uses your real entered data
Section 4 "What people did" — historical statistics	Section 4 "What people did" — demographic data for your income bracket
Simulation ends and resets	Real app never resets. Your log grows permanently.

Variables that change the real app's visual presentation:

Condition	Visual change
Burn rate under 60 days	Featured article border shifts from grey to very subtle amber
A concept you are reading is happening in real life today	Quiet margin note: "This is happening right now. Your situation: →"
Real app has no data entered yet	"Did You Know" uses national average data for your demographic
Clarity score reaches new threshold	One new "did you know" hook type becomes available on home page
A watch list item is overdue for checking	Watch list entry in margin shows subtle ! indicator

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13. Implementation Plan

Build order is strictly sequential. Each phase must be complete before the next begins. Each phase produces something usable on its own.

Phase 1 — Static Shell

Goal: The app looks complete. Both themes work. Navigation exists. Every screen exists. Nothing is functional yet.

Deliverables:

- `index.html` — full page structure with all screens as hidden divs
- `style.css` — both themes complete. `body.theme-encyclopedia` and `body.theme-newspaper` classes.
- Home page renders correctly in both themes
- Article view renders correctly (static placeholder content)
- Ghost track full view renders correctly (static)
- Clarity score screen renders correctly (static)
- Navigation sidebar/bottom bar renders correctly
- Ticker bar renders correctly (static, not scrolling yet)
- Pressure gauge renders correctly (static position)
- Theme switch works — one class change on body
- All five undeveloped sections navigate and show "no data yet" state
- iCloud folder path configured: `C:\iCloudDrive\american_pie`

Test: Open `index.html` in a browser. Switch themes. Navigate all screens. Everything looks intentional. Nothing looks broken or placeholder-like.

Phase 2 — Article Engine

Goal: The app can read a markdown article file and display it section by section.

Deliverables:

- `app.js` function: `loadArticle(filename)` — reads `.md` file from `articles/` folder
- Parser splits article into sections by `##` headers
- Section display: one section at a time, full screen, `Next →` advances
- Section counter in article header updates correctly
- Breadcrumb generates from article metadata

- Margin panel: table of contents auto-generated from section headers
- `Next →` transition: content changes in place, no scroll, no page load
- Back navigation: previous section on left swipe or back gesture
- First article loads: `articles/emergency-fund.md`

Test: Open the emergency fund article. Read all 6 sections. Advance with Next. Navigate back. Both themes look correct throughout.

Phase 3 — Variable System

Goal: Articles display real values from `variables.json`. Inline editing writes back to the file.

Deliverables:

- `app.js` function: `loadVariables(simulationYear)` — reads `variables.json`
- Template engine: finds all `{{variable_name}}` in article text and replaces with values
- Calculated variables: `coverage_days`, `net_worth` computed at load time
- Inline editor: tapping a variable value opens margin panel in edit mode
- Edit fields pre-populated with current mock values as placeholders
- On input: writes new value to `variables.json` immediately
- Article text updates inline on write — no reload
- Source view: shows file path, variables in use, calculation steps
- Mock data defaults loaded if no real data entered

Test: Open emergency fund article. See `{{your_cash}}` rendered as `$7,000`. Tap it. Edit to `$5,000`. Article updates to show new coverage days. Open source view. See the calculation.

Phase 4 — Simulation Data

Goal: The full 2008 simulation is authored and playable.

Deliverables:

- `simulations/2008/variables.json` — all 18 variables populated with historical data
- `simulations/2008/events.json` — all simulation events in sequence, one per significant moment
- `simulations/2008/ghost-tracks.json` — all five track logs from January 2008 to present

day

- `simulations/2008/did-you-know.json` — minimum 30 hooks, one per concept
- All 48 articles in the Article Manifest authored completely (see Section 10)
- Each article: all 6 sections written, all variables referenced correctly, all links between articles working
- Home page "In The News" populates from events.json for current simulation position
- Home page "Did You Know" rotates from did-you-know.json
- Ghost track margin panel reads from ghost-tracks.json at current simulation date
- Ticker populates from ghost-tracks.json current entries
- Pressure gauge reads `stress_level` from events.json and positions correctly
- Era transitions work: reaching final event of an era loads first event of next era
- Final screen displays correctly after Era 9 completes
- Reset works: simulation returns to January 2008 with no state saved

Test: Play the full 2008 simulation start to finish. Read every article. Watch the ticker. Watch the pressure gauge move. Reach the final screen. Reset. Start again.

Phase 5 — Living Components

Goal: The ticker scrolls. The pressure gauge moves. Ghost track entries update over time. The app feels alive.

Deliverables:

- Ticker: CSS scroll animation. Seamless loop. `30px/second`. Pauses on hover.
- Ticker: ghost track entries update every 30 seconds from current simulation position
- Pressure gauge: `left` CSS transition over 5000ms when stress level changes
- Ghost track margin panel: auto-refreshes when simulation advances to new event
- "On This Day" strip: rotates on each home page visit
- Navigation shell: appears after first complete article. Animates in from left (desktop) or bottom (mobile). One time only. Never shows again if already appeared.
- Ticker first appearance: renders as static illustration inside federal-reserve article. Becomes live after two more sections are read.
- Pressure gauge first appearance: renders as static illustration inside federal-reserve

article. Marker moves for first time three articles later.

- Clarity score: calculates from articles read and variables entered. Updates in real time.

Test: Open the app. Read the Federal Reserve article. Notice the illustration. Keep reading. Notice the ticker has changed. See the gauge has moved. Play through three eras. Watch the pressure gauge rise. Watch the ticker surface different events.

Phase 6 — Real App Layer

Goal: Switching to Present mode loads your real data and the real app shell works correctly.

Deliverables:

- Year filter: tapping Present switches mode. All data sources switch to `[present]/`.
- `[present]/variables.json` — template file with all variables empty. Comments explaining each field and where to find the value.
- `[present]/did-you-know.json` — hooks that calculate from real variables. Blank until variables are entered.
- `[present]/getting-started.md` — one fully authored starter article (see Section 15)
- Home page in Present mode: renders correctly with demographic average data before any variables entered
- Once first variable entered: all "did you know" hooks referencing that variable update
- Undeveloped section state: all sections except Home and Learn show "no data yet" state
- Watch list: `[present]/watch-list.json` populated with all 17 variables, sources, and check frequencies
- Printable worksheets: `worksheets/simulation-research.md` and `worksheets/real-life-snapshot.md` render correctly when printed
- `sources/where-to-find-things.md` — master reference for all data sources, fully written

Test: Switch to Present. See demographic average data. Enter your real cash amount. See the "did you know" hooks update. Tap the emergency fund article. See it load. Enter your real monthly expenses. See your real coverage days calculate. Open source view. See your real file path.

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14. iCloud Sync Setup

Your iCloud folder path is `C:\iCloudDrive\american_pie` **.**

```
<details>
<summary><strong>Step 1 — Verify iCloud Drive is installed and syncing</strong></summar
```

On Windows: Open File Explorer. Look for iCloud Drive in the left panel. If not present, download iCloud for Windows from apple.com/icloud/icloud-for-windows/.

Sign in with your Apple ID. Enable iCloud Drive sync. Verify the folder appears at `C:\iCloudDrive\` **.**

```
</details>
```

```
<details>
<summary><strong>Step 2 — Create the project folder</strong></summary>
```

Navigate to `C:\iCloudDrive\` **. Create a new folder named** `american_pie` **. This is your project root. Everything lives here.**

```
</details>
```

```
<details>
<summary><strong>Step 3 — Open the app</strong></summary>
```

Place `index.html` **in** `C:\iCloudDrive\american_pie\` **. Open it in any browser. The app configures its file paths relative to this location automatically.**

```
</details>
```

```
<details>
<summary><strong>Step 4 — Access on mobile</strong></summary>
```

On your iPhone or iPad: open the Files app. Tap iCloud Drive. Navigate to `american_pie` **. Tap** `index.html` **. It opens in Safari. Tap the share icon and select "Add to Home Screen" to create an app icon. It behaves like a native app.**

```
</details>
```

```
<details>
<summary><strong>Step 5 — Sync behavior</strong></summary>
```

Any file you edit on desktop syncs to iCloud and becomes available on mobile within seconds

on a good connection. Any variable you enter on your phone syncs to your laptop. There is no manual sync step. iCloud handles it.

If you are offline: the app reads the last cached version of all files. Changes you make while offline queue and sync when reconnected.

```
</details>


<details>
<summary><strong>Step 6 — Backend editing on desktop</strong></summary>
```

Any text editor works for editing `.json` and `.md` files. Recommended: Visual Studio Code (free). Install it. Open the `american_pie` folder as a workspace. You can edit any file, see the folder structure, and use the built-in terminal — all from one window.

This is what "editing the backend" looks like. A folder of text files in a text editor. The same thing the app's UI does when you tap an editable field.

```
</details>
```

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15. First Article — Complete Authored Example

This article is fully written and ready to place at `articles/emergency-fund.md`. It serves as the template for every article that follows.

```
---
title: Emergency Fund
era: 1
date: March 2008
category: Banking
sections: 6
concepts: [emergency-fund, burn-rate, coverage-days]
related: [burn-rate, checking-vs-savings, unemployment-insurance]
---

## What It Is

An emergency fund is money set aside specifically to cover
unexpected expenses or loss of income without disrupting your
regular financial life. It is not an investment. It is not
savings toward a goal. It is a buffer between you and the
decisions people make when they are desperate.
```

In March 2008, the median American household had $6,800 in liquid savings. This sounds like a cushion. For most households, it covered less than 47 days of expenses. Many did not know that number until they needed it.

Your estimated coverage: **{{coverage_days}} days**

Your Number

Your emergency fund coverage is calculated from two numbers you already know: how much cash you have and how much you spend each month.

Monthly expenses: **{{your_monthly_expenses}}**
Current cash: **{{your_cash}}**

Coverage: {{your_cash}} ÷ {{your_monthly_expenses}} × 30
= **{{coverage_days}} days**

The standard recommendation is 90 to 180 days. Most financial advisors consider anything under 30 days a critical gap. Not because catastrophe is certain, but because you are one event away from choices you do not want to make.

The Math

Why does 90 days matter specifically?

The average job search in 2008 took 5 months for someone with marketable skills in a normal economy. In a recession, that extended to 7 to 9 months. Unemployment insurance, if applied for immediately, replaces approximately 40 to 50 percent of prior income. The gap between what unemployment pays and what you actually spend must come from somewhere.

At {{your_monthly_expenses}} per month with unemployment covering 45 percent:

Monthly gap: {{your_monthly_expenses}} × 0.55
= approximately **{{income_gap}}**

At that gap rate, your {{your_cash}} covers approximately **{{covered_months_with_ui}} months** with unemployment assistance, or **{{coverage_days}} days** without it.

This is your number. Knowing it is the entire point of this section.

What People Did

What most people did

In 2008, 61 percent of Americans who lost their job waited more than three weeks to reduce spending. By then, the average household had used $1,400 of savings unnecessarily. Most did not calculate their burn rate until the third or fourth week of unemployment. By then, the decisions became harder.

The second most common mistake: withdrawing from a 401k. Early withdrawal costs 10 percent penalty plus income tax on the amount withdrawn. A $10,000 withdrawal might net $6,500 after penalties and taxes. Most people who did this in 2008 called it the most expensive mistake of their financial lives.

What financially stable people did

They calculated their exact burn rate within 48 hours of job loss and cut non-essential spending immediately. Not because they panicked. Because the math was clear. They knew their number. They knew how long they had. They did not guess.

They also filed for unemployment insurance immediately. Not three weeks later. Immediately. The first check takes three to four weeks to arrive regardless of when you file. Waiting to file means waiting longer for the first check.

▸ If you had known this before 2008

The average person who calculated their burn rate and filed for unemployment within 48 hours of job loss in 2008 preserved $2,300 more of their savings over the following six months than someone who waited. That $2,300, invested in March 2009 during the market bottom, would be worth approximately $15,700 by 2024.

The lesson is not about the money. The lesson is that knowing your number in advance removes the paralysis that causes the delay that causes the unnecessary spending.

What It Connects To

This concept touches three others directly:

→ **Burn Rate** — the monthly expense number that powers all emergency fund calculations. Knowing your burn rate

```
  precisely makes this number real instead of estimated.

→ **Checking vs Savings** — where your emergency fund
  lives matters. Cash sitting in checking earns nothing.
  The same cash in a high-yield savings account earned
  4.5 percent in early 2008. On $7,000 that is $315 per
  year for doing nothing differently.

→ **Unemployment Insurance** — how UI payments interact
  with your emergency fund changes the calculation
  significantly. Understanding both together gives you a
  far more accurate picture of your actual runway.

## Your Next Step

Find your last three months of bank statements. Add the
total spending for each month. Divide by three. That is
your burn rate. Divide your current cash by that number
and multiply by 30.

That is how many days you have. Write it down. It is the
most important number in your financial life right now and
most people have never calculated it.

If you are using mock data, your number is {{coverage_days}}
days. To see your real number, tap any value above and
enter your actual figures.
```

Note on the real app starter article:

Place the following at `articles/[present]/getting-started.md`. This is the one article that exists in the real app on day one. It uses real variables. It teaches you what to enter first and why.

```
---
title: Getting Started With Your Real Numbers
era: present
date: {{today}}
category: Your Situation
sections: 4
concepts: [real-data, first-steps, document-organization]
---

## The First Number That Matters

Before anything else in this app becomes personal, you
need one number: your monthly burn rate.
```

This is not your income. It is not your rent. It is the total amount of money you spend in an average month on everything. Rent, food, transportation, subscriptions, utilities, everything.

You do not need to know this precisely to start. An estimate within $200 is close enough for now. You will refine it as you use the app.

To find it: open your last three bank statements. Add the total money that left your account each month. Divide by three. That is your burn rate.

Tap the value below to enter your number:

Your monthly burn rate: **{{your_monthly_expenses}}**

What To Enter Next

With your burn rate entered, two things become real:

Your coverage days with current cash:
{{coverage_days}} days

Your monthly income gap if you lost income today:
{{income_gap}} (what expenses exceed unemployment)

These two numbers determine everything else in this app. Every article, every calculation, every "did you know" item will become personal once these are set.

Enter your current cash next. It is on your bank app right now. Just the checking and savings total. Not investments yet.

Your current cash: **{{your_cash}}**

How To Organize Your Documents

You do not need to upload anything. You do not need to scan anything. You need to know where things are when the app asks for a specific number.

Create one folder on your computer:

Documents/ └── finances/ ├── bank-statements/ ├── pay-stubs/ ├── tax-returns/ └── other/

```
When the app asks for a number, it will tell you exactly
which document has it and exactly where on the document
to look. Your job is to have the document in the right
folder so you can find it in under 30 seconds.

That is the entire organization system. Simple enough
to actually use. Structured enough to actually work.

## Your First Watch List Item

The first thing to monitor in your real financial life
is the Federal Reserve interest rate. Not because you
are investing. Because the rate affects your savings
account, any debt you carry, and the job market you
are navigating.

Check it once a month at:
federalreserve.gov → Monetary Policy → Target Rate

Current rate as you have entered it: **{{fed_funds_rate}}**

When it changes, update that value in your variables
file. The app will show you exactly what the change
means for your specific situation.
```

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